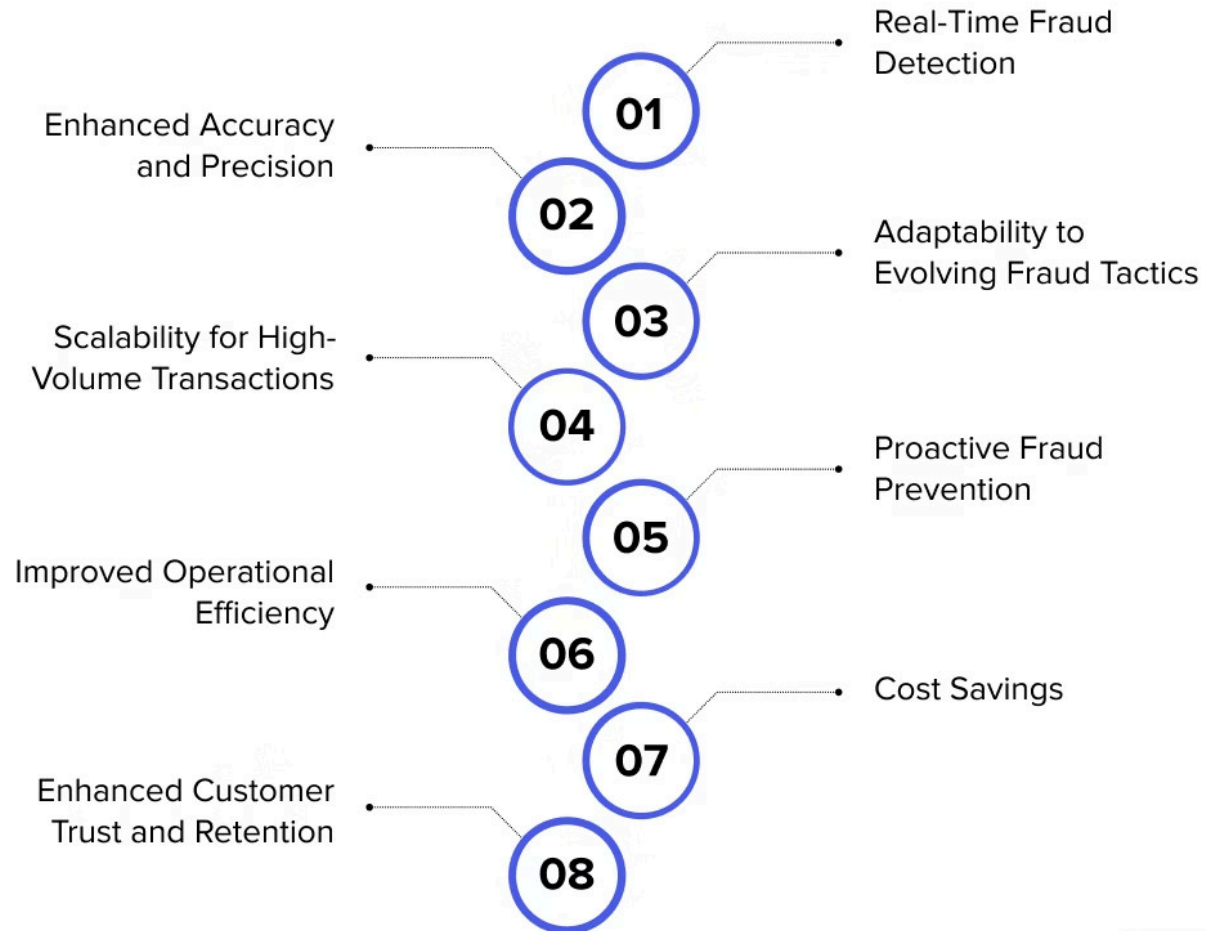


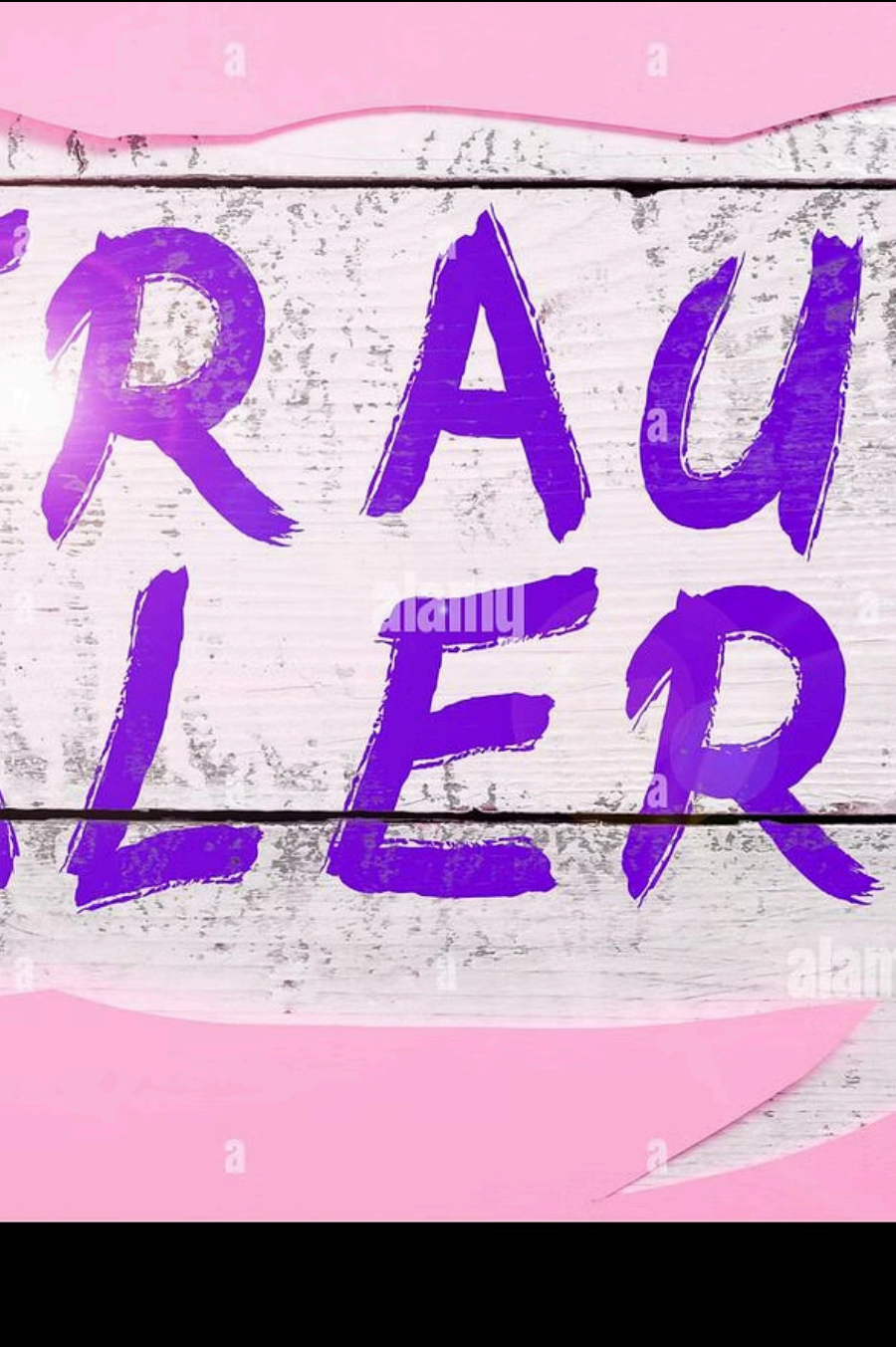
Benefits of Machine Learning in Credit Card Fraud Prevention



Credit Card Fraud Detection Using Machine Learning

AI23331 – Fundamentals of Machine Learning

Student Name(s) · Register Number(s) · Date



Introduction



What Is Credit Card Fraud?

Unauthorized use of card details to make transactions without the cardholder's consent.



A Serious Global Problem

Fraud costs billions annually, erodes consumer trust, and grows with digital payment adoption.



Why Detection Matters

Real-time ML detection minimizes financial loss and protects users before damage spreads.

Problem Statement



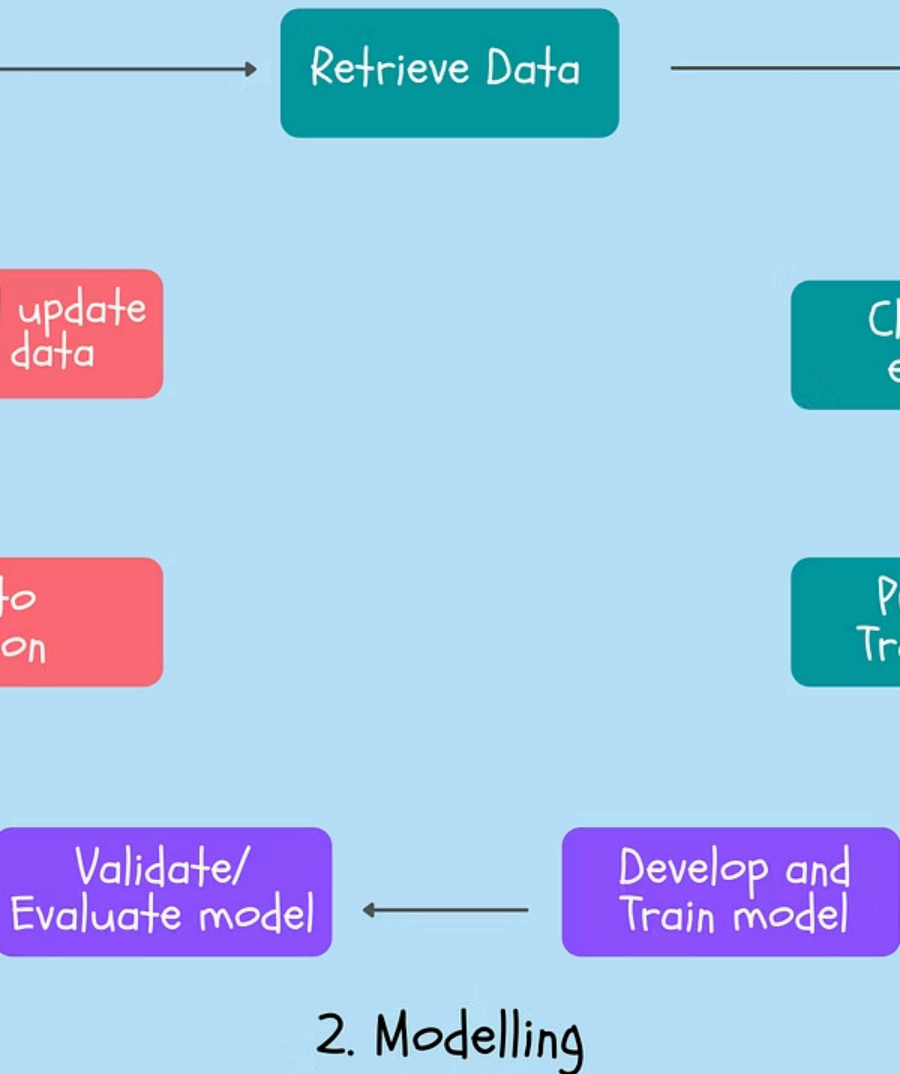
Severe Class Imbalance

Fraud accounts for only **~0.17%** of transactions – extreme imbalance challenges standard classifiers.

Why Traditional Models Fail

- High accuracy is misleading when fraud is rare
- Models default to predicting "legitimate" for all cases
- Cost of false negatives is extremely high

Machine Learning Workflow



Project Objectives



Data Preprocessing

Clean, scale, and prepare transaction data



Handle Imbalance

Apply SMOTE to balance fraud vs. legitimate classes



Model & Evaluate

Train multiple models; evaluate with AUPRC, F1, ROC-AUC



Deploy

Serve predictions via FastAPI REST endpoint

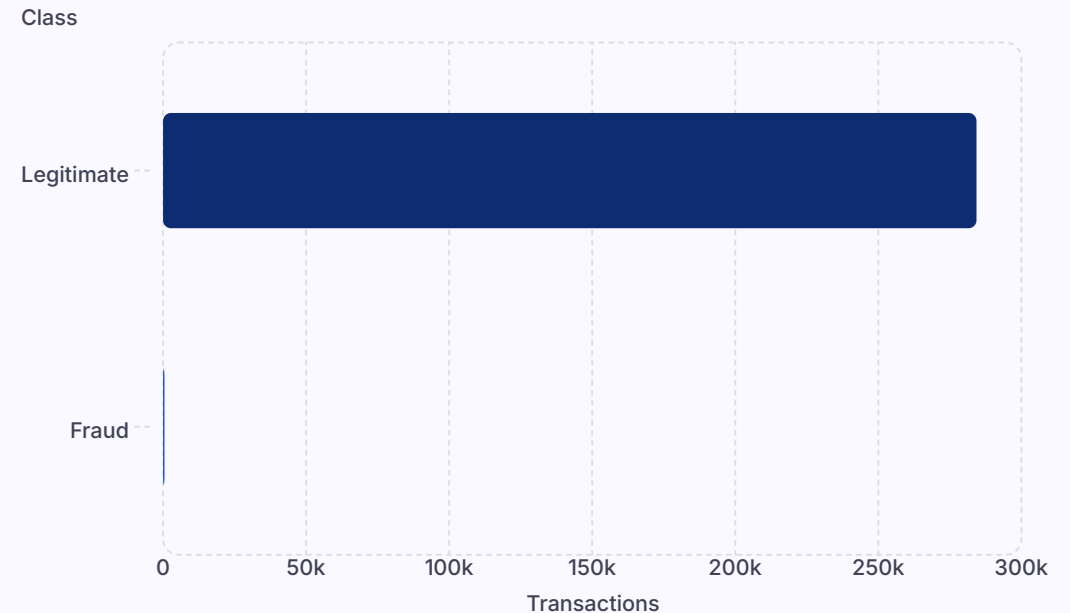
Dataset & Exploratory Data Analysis

Dataset Overview

- Kaggle European cardholders dataset
- 284,807 transactions over 2 days
- 30 features: Time, Amount, V1 - V28 (PCA)
- Class: 0 = Legitimate, 1 = Fraud

Key Insights

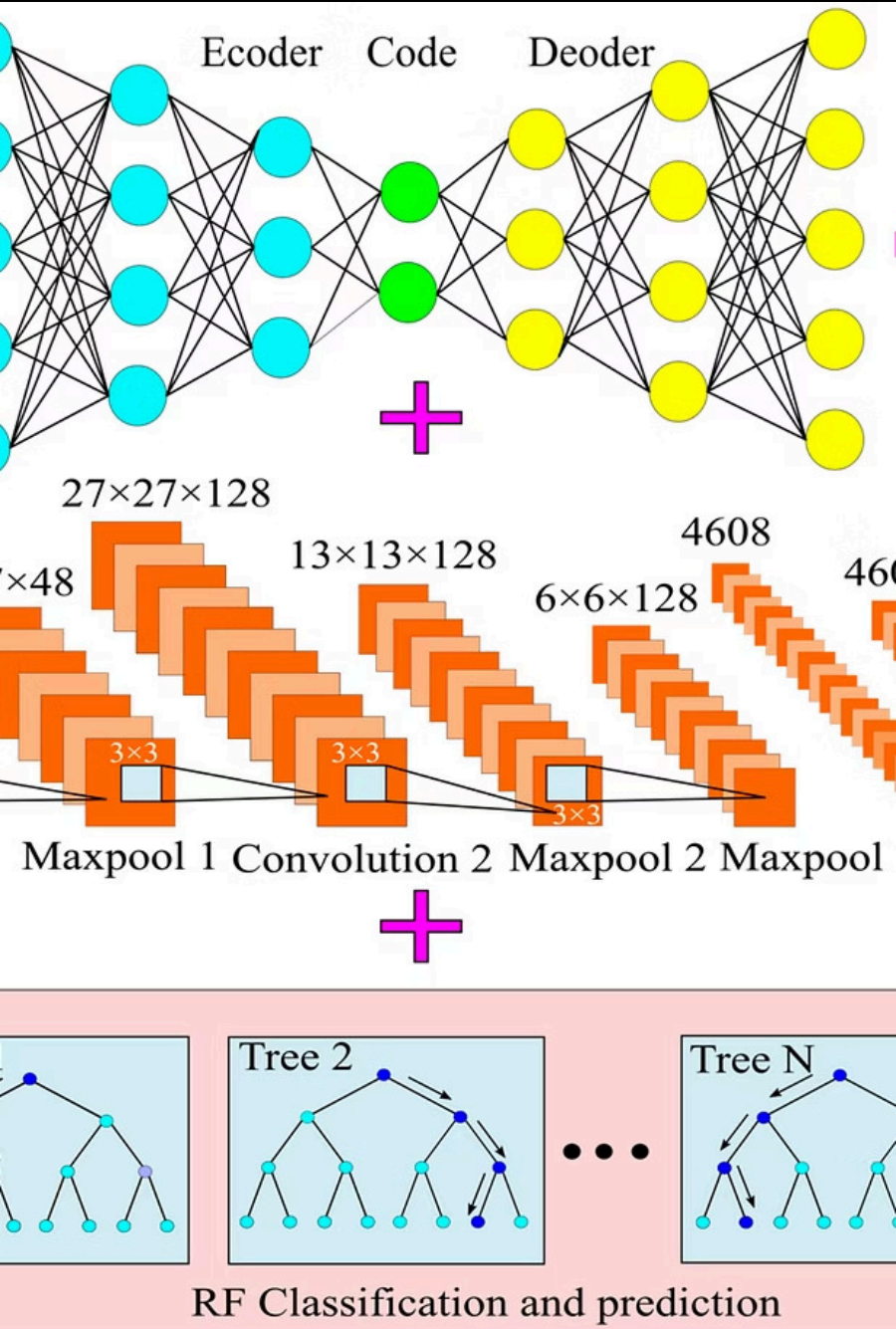
- Fraud rate: **~0.17%**
- Fraudulent transactions tend to be smaller amounts
- PCA features preserve privacy while retaining signal



Methodology



SMOTE (Synthetic Minority Over-sampling Technique) generates synthetic fraud samples to address class imbalance before training, ensuring models learn fraud patterns rather than defaulting to majority class.



Models Used



Logistic Regression

Baseline model – interpretable, fast, linear decision boundary



Random Forest

Ensemble of decision trees – handles non-linearity well



XGBoost

Final model – gradient boosting, high accuracy, handles imbalance



Isolation Forest

Unsupervised anomaly detection – isolates outliers efficiently

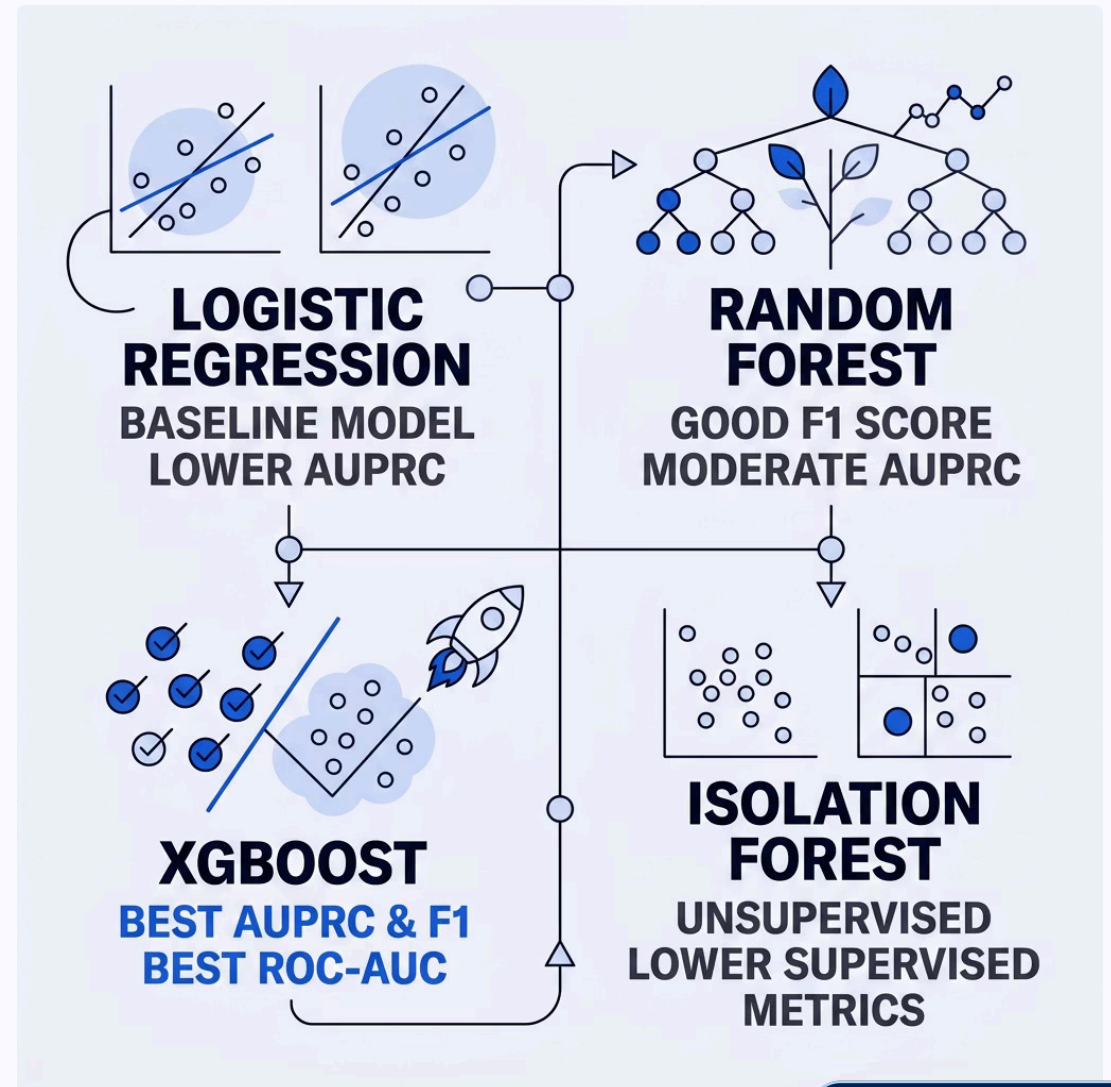
Results & Evaluation

Why AUPRC?

With 0.17% fraud, accuracy is meaningless. **AUPRC** focuses on the minority class – the true measure of fraud detection quality.

Best Model: XGBoost

- Highest AUPRC and F1 Score across all models
- Strong ROC-AUC confirming discriminative power
- Outperformed Logistic Regression, Random Forest, and Isolation Forest



Deployment



Serving the Model

- Model serialized and saved using **Joblib**
- **FastAPI** provides a lightweight REST API for real-time predictions
- Accepts transaction features, returns fraud probability score
- Low latency – suitable for production fraud screening

i API endpoint: `POST /predict` → returns `{"fraud": true/false, "score": 0.98}`

Conclusion & Future Work

Summary

XGBoost achieved the best fraud detection performance using AUPRC and F1 Score on a highly imbalanced dataset.

Deep Learning

Explore autoencoders and LSTM networks for sequential transaction patterns.

SHAP Explainability

Add interpretable feature importance to explain predictions to stakeholders.

Real-Time Systems

Integrate streaming pipelines (Kafka, Spark) for live fraud monitoring at scale.